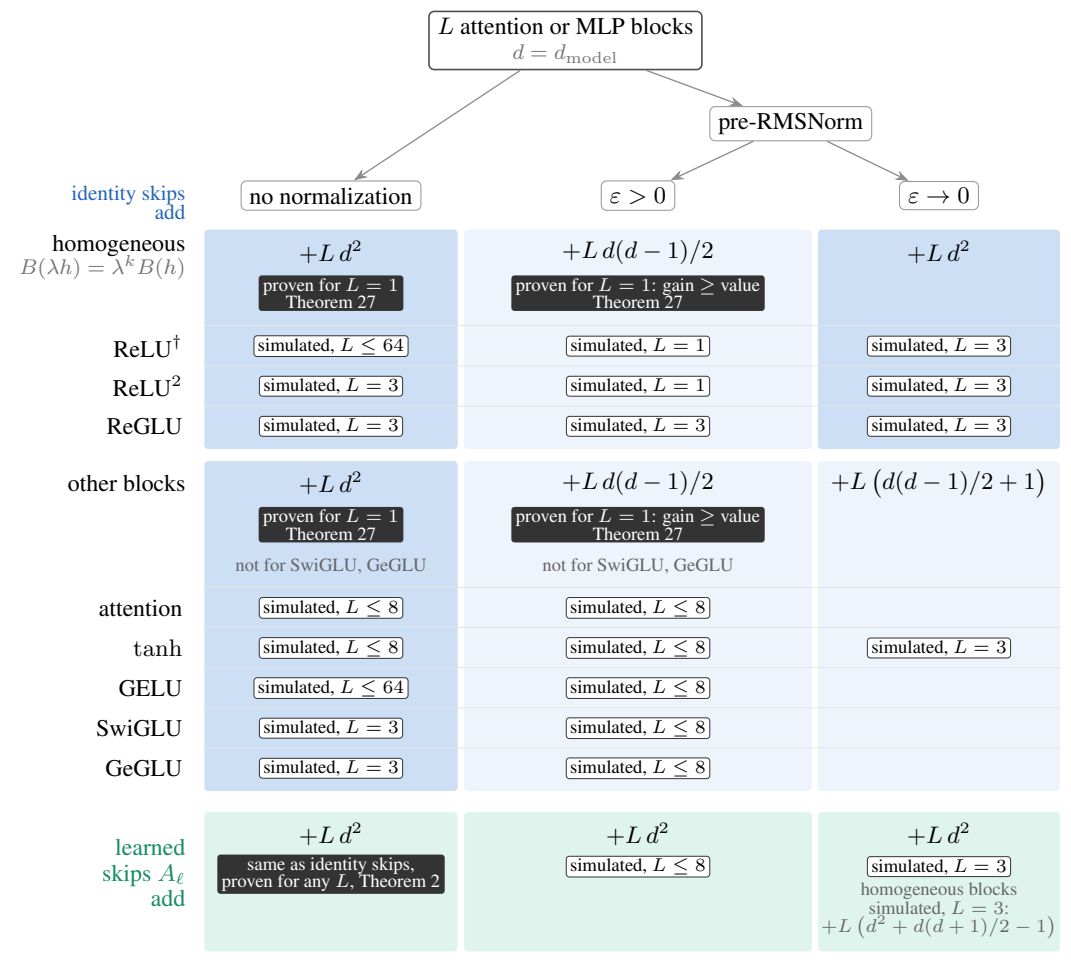

$L$ attention or MLP blocks
$d = d_{\text{model}}$
pre-RMSNorm
identity skips add
no normalization
$\varepsilon > 0$
$\varepsilon \to 0$
homogeneous $B(\lambda h) = \lambda^k B(h)$
$+L\,d^2$
proven for $L = 1$ Theorem 27
$+L\,d(d-1)/2$
proven for $L = 1$: gain $\geq$ value Theorem 27
$+L\,d^2$
ReLU$^\dagger$
simulated, $L \leq 64$
simulated, $L = 1$
simulated, $L = 3$
ReLU$^2$
simulated, $L = 3$
simulated, $L = 1$
simulated, $L = 3$
ReGLU
simulated, $L = 3$
simulated, $L = 3$
simulated, $L = 3$
other blocks
$+L\,d^2$
proven for $L = 1$ Theorem 27
not for SwiGLU, GeGLU
$+L\,d(d-1)/2$
proven for $L = 1$: gain $\geq$ value Theorem 27
not for SwiGLU, GeGLU
$+L\left(d(d-1)/2+1\right)$
attention
simulated, $L \leq 8$
simulated, $L \leq 8$
tanh
simulated, $L \leq 8$
simulated, $L \leq 8$
simulated, $L = 3$
GELU
simulated, $L \leq 64$
simulated, $L \leq 8$
SwiGLU
simulated, $L = 3$
simulated, $L \leq 8$
GeGLU
simulated, $L = 3$
simulated, $L \leq 8$
learned skips $A_\ell$ add
$+L\,d^2$
same as identity skips, proven for any $L$, Theorem 2
$+L\,d^2$
simulated, $L \leq 8$
$+L\,d^2$
simulated, $L = 3$
homogeneous blocks simulated, $L = 3$:
$+L\left(d^2 + d(d+1)/2 - 1\right)$